Alan Muniz

Assistant Professor

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Personal information

Place and date of birth: Rio de Janeiro, Brazil; April 2nd, 1987.
Professional address: Departamento de Matemática, Centro de Ciências Exatas e da Natureza, Universidade Federal de Pernambuco (UFPE) – Recife, BR

ORCID iD

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Education

2012–2016 **Ph.D. in Mathematics (Algebraic Geometry and Holomorphic Foliations)**, *Universidade Federal de Minas Gerais (UFMG), February 25th, 2016*, Belo Horizonte
Thesis on: *"Polynomial Bounds for automorphism groups of foliations"*
Advisor: Prof. *Maurício Corrêa*.

2010–2012 **Master Degree in Mathematics**, *Universidade Federal Fluminense (UFF), February 9th, 2012*
Thesis on: *"Integrais primeiras de folheações holomorfas"*
Advisor: Prof. *Thiago Fassarella*.

2006–2010 **Full Licentiate Degree in Mathematics**, *Universidade Estadual do Rio de Janeiro, January 31st, 2010*,

Present Position

2024– **Professor Assistente**, *Universidade Federal de Pernambuco*, Recife, Brazil

Past Employment

2022–2023 **Postdoc position**, *Universidade Estadual de Campinas*, Campinas, Brazil
Suppervised by Marcos Jardim

2021–2022 **Postdoc position**, *Université de Bourgogne*, Dijon, France
Suppervised by Daniele Faenzi

2018–2019 **Postdoc position**, *Universidade Federal do Espírito Santo*, Vitória, Brazil
Supervised by Leonardo Câmara

2016–2018 **Postdoc position**, *Universidade Federal Fluminense*, Niterói, Brazil
Suppervised by Thiago Fassarella

2017–2018 **Temporary professor**, *Universidade Federal Fluminense*

Awards and Honours

- CNPq Research Productivity Fellowship, Level C, CNPq grant number 305599/2026-7
- INCTMat Postdoctoral Fellowship, CNPq grant number 160934/2022-2.
- CAPES-COFECUB Postdoctoral Fellowship 88887.468856/2019-00, under the project number

88887.191919/2018-00.
- PNPD/CAPES Postdoctoral Fellowship, 11/2018-10/2019.
- PNPD/CAPES Postdoctoral Fellowship, 11/2016-10/2018.
- CAPES-REUNI Doctorate Program Scholarship, 03/2012-02/2016.
- CAPES Master's Program Scholarship, 03/2010-02/2012.

Publications

1. V. Benedetti, C. Kuster, and A.Muniz, *Minimal-degree foliations on cominuscule Grassmannians.* **Bulletin of the Brazilian Mathematical Society**, *to appear*.
2. M Figueira, C. Kuster, R. Lizarbe, and A.Muniz, *Degree-one foliations on complete intersections.* **Journal of Pure and Applied Algebra**, 2026.
3. K. Kozhasov, A. Muniz, Y. Qi, and L. Sodomaco, *On the minimal algebraic complexity of the rank-one approximation problem for general inner products.* **Mathematics of Computation**, 2025.
4. M. Jardim and A. Muniz, *Rank-two reflexive sheaves on the projective space with second Chern class equal to four.* **Bulletin of the Brazilian Mathematical Society**, 2025.
5. A. Muniz, *p-Forms from Syzygies.* **Rendiconti dell'Istituto di Matematica dell'Università di Trieste**, 2025.
6. M. Corrêa, M. Jardim, A. Muniz, *Moduli of Distributions via Singular Schemes.* **Mathematische Zeitschrift**, 2022.
7. V. Benedetti, D. Faenzi, and A. Muniz, *Codimension one foliations on homogeneous varieties.* **Advances in Mathematics**, 2023.
8. T. Fassarella, F. Loray, and A. Muniz, *On the Moduli of Logarithmic Connections on Elliptic Curves.*, **Mathematische Zeitschrift**, 2022.
9. H. Galeano, M. Jardim, and A. Muniz, *Codimension one distributions of degree 2 on the three-dimensional projective space.* **Journal of Pure and Applied Algebra**, 2022.
10. A. Muniz and R. Rosas, *Foliations on the projective plane with finite group of symmetries.* **Annali della Scuola Normale Superiore di Pisa, Classe di Scienze**, 2021.
11. M. Corrêa and A. Muniz *Polynomial bounds for automorphisms groups of foliations.* **Revista Matemática Iberoamericana**, 2019.

Summary of Publications

- **Number of publications:** 9
- **Number of citations:** 14
- **H-index:** 2 (WebOfScience)
- **Main works:** From the list above: 1,4,5,6,7.

Experience as a supervisor

Ph.D. students:

2026– **Ph.D. Thesis**, *Mateus Ferreira de Melo*, Decidability in the class of tame henselian fields, Universidade Federal de Pernambuco

2022– **Ph.D. Thesis**, *Victor Zaher*, Folheações por curvas no espaço projetivo., Universidade Estadual de Campinas-UNICAMP, Campinas
Co-advised with Marcos Jardim

2024– **Ph.D. Thesis**, *Pedro Pfarrius Barbassa*, Moduli spaces of distributions on projective spaces, Universidade Estadual de Campinas-UNICAMP, Campinas
Co-advised with Marcos Jardim

Master students:

2026– **Master Thesis**, *Gabriel Luz Pachêco*, Estruturas de Engel em Variedades Complexas: Análise Local e Global, Universidade Federal de Pernambuco

2023–2024 **Master Thesis**, *Pedro Pfarrius Barbassa*, Esquema singular de folheações por curvas., Universidade Estadual de Campinas-UNICAMP, Campinas
Co-advised with Marcos Jardim

Invited lectures

- *Distribuições em Espaços Projetivos*, Programa de Verão 2024, Universidade Federal Fluminense (UFF), Niterói, Brazil. 2024.

Talks as invited speaker in conferences and seminars

- *Foliations on Complete Intersections*, Foliations, Complex Geometry, and Painlevé Equations, Instituto de Matemática Pura e Aplicada (IMPA), Rio de Janeiro-Brazil. 2025
- *Foliations on Homogeneous Spaces*, From Schubert Calculus to Representation Theory, Universidade Federal de Sergipe (UFS), Aracaju-Brazil. 2024.
- *Distribution and syzygies*, Bandoleros 2023, Università di Torino, Torino-Italy. 2023.
- *On the moduli space of rank two logarithmic connections over an elliptic curve*, Seminaire de Géométrie Analytique, Université de Rennes, Rennes-France. 2022.
- *Codimension one Foliations on Homogeneous Spaces*, Séminaire de Géométrie Algébrique, Université de Lille, Lille-France. 2022.
- *Moduli of Distributions via Singular Schemes*, Seminari di Matematica, Università di Bari, Bari-Italy. 2022
- *Simetrias de foliaciones en el plano proyectivo*, III Congreso Internacional de Geometría y Dinámica Compleja, Pontificia Universidad Católica del Perú, Lima-Peru. 2018.

Scientific Committees

- "3rd International Meeting of Young Researchers in Singularity Theory and Related Fields" – 2024.
- "SAGüi - Seminário de Álgebra e Geometria do DMat-UFPE" (co-organizer) – 2024-Present.

Evaluation activities

- Evaluating research projects for FACEPE, financing research projects based in Pernambuco, Brazil. 2024- Present.
- Evaluating research projects for CNPq, Brazil. 2026- Present.